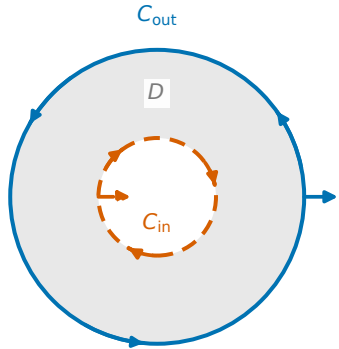
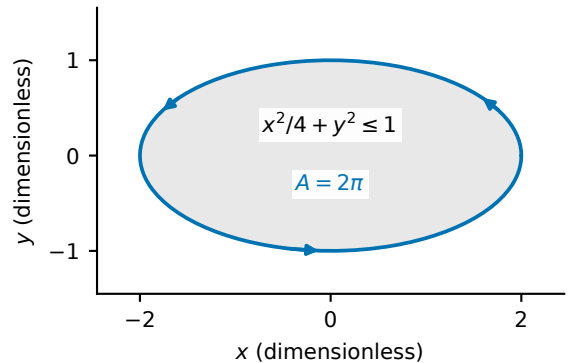
A



$\partial D = C_{\text{out}} + C_{\text{in}}$

B



$A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$